Weichi Yao

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Education

2026- **Postdoctoral Research Fellow**, Statistics Department - University of Michigan

AI Mentor Prof. Yixin Wang
Science Mentor Prof. Bryan R. Goldsmith

2023-2025 **Postdoctoral Research Fellow**, Schmidt AI in Science - University of Michigan

AI Mentor Prof. Yixin Wang
Science Mentor Prof. Bryan R. Goldsmith

2017-2023 **Ph.D.**, Statistics - New York University
M.Phil, Statistics

Advisor Prof. Halina Frydman
Thesis: Machine Learning for Science

2015-2016 **M.A.**, Statistics - Columbia University

2011-2015 **B.A.**, Finance - Wuhan University
B.A., Applied Mathematics

Grants

2025-2027 ACED: Tail-aware Generative Modeling for Inverse Discovery of Molecules (Co-PI).

National Science Foundation (Total: $500K).

PI: Y. Wang (Statistics); Co-PI: B. R. Goldsmith (Chemical Engineering).

2025-2026 Reasoning-Augmented Large Language Models for Molecule Design and Discovery.

University of Michigan-OpenAI Research Grants (Total: $50K).

PI: B. R. Goldsmith (Chemical Engineering).

Ongoing Projects

1. W. Yao, B. R. Goldsmith and Y. Wang.

Extrapolative conditional molecule generation.

Key contribution: Building a lightweight transformer adapter for pretrained conditional generative model, enabling reliable generation when target properties outside the training distribution without retraining the base model. Validated the method on synthetic benchmarks and QM9 molecular inverse-design tasks, improving out-of-distribution accuracy and extrapolative robustness over the baseline generator.

2. D. Sweeney, W. Yao, Y. Wang and B. R. Goldsmith

A conditional flow matching model for heterogeneous catalyst design.

Key contribution: Developing a flow matching model with transformer as backbone, which generates complete catalyst–adsorbate configurations from scratch, including catalyst structures, adsorbate atom types, and optimized adsorption geometries, conditioned on target binding energies.

Preprints

2026 W. Yao, C. Gruich, Y. Wang and B. R. Goldsmith. Fixed-Dimensional Latent Flow for Generating Variable-Size 3D Molecules. *arXiv preprint: 2609.08333*.

2026 C. Gruich, W. Yao, Y. Wang and B. R. Goldsmith. Adapting Evidential Neural Networks to Test-Time Neighbor Fusion Improves Molecular Property Prediction. *arXiv preprint: 2607.11091*.

2026 W. Yao, B. Dumitrascu, B. R. Goldsmith and Y. Wang. Goal-Oriented Influence-Maximizing Data Acquisition for Learning and Optimization. *arXiv preprint: 2602.19578*.

2024 W. Yao, C. Gruich, B. R. Goldsmith and Y. Wang. Tail extrapolative conditional molecule generation. *In Proceedings of the International Conference on Machine Learning (ICML) AI4Science Workshop*.

2021 W. Yao, K. Storey-Fisher, D. W. Hogg and S. Villar. A simple equivariant machine learning method for dynamics based on scalars. *In Proceedings of the Advances in Neural Information* Processing Systems *(NeurIPS) Workshop on Machine Learning and the Physical Sciences*.

Publications

2024 S. Villar, D. W. Hogg, W. Yao, G. A. Kevrekidis and B. Schölkopf. Towards fully covariant machine learning. *Transactions on Machine Learning Research*.

2023 S. Villar, W. Yao, D. W. Hogg, B. Blum-Smith and B. Dumitrascu. Dimensionless machine learning: Imposing exact units equivariance. *Journal of Machine Learning Research, 24*.

2022 W. Yao, H. Frydman, D. Larocque and J. S. Simonoff. Ensemble methods for survival function estimation with time-varying covariates. *Statistical Methods in Medical Research*, 31(11): 2217-2236.

2021 W. Yao, H. Frydman and J. S. Simonoff. An ensemble method for interval-censored time-to-event data. *Biostatistics*, 22(1): 198-213.

2021 H. Moradian, W. Yao, D. Larocque, J. S. Simonoff and H. Frydman. Dynamic estimation with random forests for discrete-time survival data. *The Canadian Journal of Statistics*, 50(2): 533-548.

2021 S. Villar, D. W. Hogg, K. Storey-Fisher, W. Yao and B. Blum-Smith. Scalars are universal: Equivariant machine learning, structured like classical physics. *In Proceedings of the Advances in Neural Information Processing Systems*

2019 W. Yao, A. S. Bandeira and S. Villar. Experimental performance of graph neural networks on random instances of max-cut. *In Proceedings of the Society of Photographic Instrumentation Engineers.*

2017 J. H. Lee, D. E. Carlson, H. S. Razaghi, W. Yao, G. A. Goetz, E. Hagen, E. Batty, E. J. Chichilnisky, G. T. Einevoll and L. Paninski. YASS: Yet Another Spike Sorter. *In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS)*, 4005-4015.

Internship

12/2022-05/2023 Science Intern. Amazon Development Center, Germany
Transfer learning with deep tabular models.

05/2022-09/2022 Research Intern. Applied Machine Learning Team, TikTok at ByteDance
Maximum likelihood estimation with full-likelihood formulation for click-through rate prediction on nonuniform subsampled data.

12/2016-07/2017 Research Assistant. Grossman Center for the Statistics of Mind, Columbia University
Statistical models and pipelines for spike sorting on dense multi-electrode arrays.

Teaching experience

2020 Course Instructor. STAT-UB Statistics for Business Control, New York University.

Prizes or Awards

2014 Honorable Mention. Mathematical Contest in Modeling (MCM)

2013 Second Prize. Symphony Orchestra, National College Students Art Exhibition, China.

2009 Gold Award. Symphony Orchestra, Australian International Music Festival, Australia.